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Collision cross sections and electron swarm parameters in argon

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Abstract. The electron distribution function is calculated in argon by solving the Boltzmann equation in the classical two-term approximation for reduced electric fields in the range $10^{-19} \leq E/N \leq 2 \times 10^{-15} \text{ V cm}^2$. A detailed set of collision cross sections including about 30 excitation levels is used. This set of inelastic cross sections is obtained directly from a microscopic approach without any adjustment to fit the calculated and measured electron swarm parameters. Our predicted values of these parameters are in excellent agreement with the most recent experimental data. For each level, the excitation rate coefficients are reported for the direct excitation from the ground state and for the cascades from upper states. They agree with the experimental results when the latter are available, but for most levels our results are the first ones.

1. Introduction

In discharge-pumped lasers, the electron–atom collisions determine essential properties such as the kinetic pathways leading to laser emission or the development of instabilities which can severely limit the performances and the scalability of the system. In many lasers, rare gases play a major role and a detailed investigation of the excitation rate coefficients of their different levels appears fruitful, especially for systems like the rare-gas halide or metal vapour rare-gas lasers in which the ionisation does not result from single-step processes but rather, arises via multi-step channels. Up to now, a realistic modelling of these lasers has suffered from the absence of a complete and self-consistent set of electron impact cross sections.

Recently, we have proposed (Bretagne *et al* 1986, hereafter referred to as I) a set of electron impact inelastic collision cross sections based upon the Bethe–Born formalism, some internal consistency checks and the available experimental data; among these we have extensively used the recent differential cross section measurements by Chutjian and Cartwright (1981). This set of cross sections proved very efficient in the analysis of the plasmas produced by relativistic electron beams (REB). Of course, in these works devoted to the study of the interaction between matter and high-energy incident particles, the attention was essentially focused on high-energy processes. The basic aim of the present paper is to extend the analysis to the low-energy range and to calculate a detailed set of argon excitation rate coefficients usable in the modelling of discharge-

pumped lasers. Fortunately, for the first time, an experimental investigation of the excitation rate coefficients of some individual levels of argon has been reported (Tachibana 1985) and can be used as a comparison with the prediction of the present calculations.

2. Theoretical formulation

To calculate the excitation rate coefficients as well as the transport parameters, firstly one must compute the electron distribution function. For that purpose, we use an appropriate code which solves the Boltzmann equation in the usual two-term spherical harmonic expansion. As the general formulation has been previously given elsewhere (Brunet and Vincent 1979), we indicate here only the basic expressions. In a spatially dependent steady state approximation, the Boltzmann equation for the electrons takes the following integro-differential form:

$$\begin{aligned} \frac{E^2}{3N^2} \frac{u}{Q_m} \frac{df}{du} + \frac{2m}{M} u^2 Q_m \left(f + \frac{kT}{e} \frac{df}{du} \right) + \frac{\alpha_i E}{3N^2} \frac{uf}{Q_m} + \frac{\alpha_i E}{3N^2} \int_0^u \frac{u}{Q_m} \frac{df}{du} du + \frac{\alpha_i^2}{3N^2} \int_0^u \frac{uf}{Q_m} du \\ = \sum_{j \neq i} \int_u^{u+u_j} u Q_j(u) f(u) du - \int_0^u du \int_{u+u_i}^{2u+u_i} z q_i(z, z-u-u_i) f(z) dz \\ - \int_0^u du \int_{2u+u_i}^\infty z q_i(z, u) f(z) dz + \int_0^u u Q_i(u) f(u) du \end{aligned} \quad (1)$$

where E is the electric field, N is the gas density, M is the mass of the molecule, k is the Boltzmann constant and $u = mv^2/2e$ is the electron energy in volts. Q_m is the electron momentum-transfer cross section, Q_j is the inelastic cross section for the excitation of the j th level, $q_i(u, u')$ is the differential ionisation cross section and f is the isotropic part of the electron distribution function.

Once the distribution function has been calculated for a given E/N , the swarm parameters and the excitation rate coefficients are obtained through the following relations:

$$v_d = - \left(\frac{2e}{m} \right)^{1/2} \frac{E}{3N} \int_0^\infty \frac{u}{Q_m} \frac{df}{du} du \quad (2)$$

$$D_r = \left(\frac{2e}{m} \right)^{1/2} \frac{1}{3N} \int_0^\infty \frac{u}{Q_m} f du \quad (3)$$

$$k_j = \left(\frac{2e}{m} \right)^{1/2} \int_0^\infty Q_j(u) f(u) du \quad (4)$$

where v_d is the drift velocity, D_r the radial diffusion coefficient and k_j the excitation rate coefficient for the j th inelastic process. For ionisation, the cross section $Q_i(u)$ appearing in (4) is the total ionisation cross section which is related to $q_i(u, u')$ through:

$$Q_i(u) = \int_0^{(u-I)/2} q_i(u, u') du' \quad (5)$$

where I is the first ionisation potential of argon. All the above relations stand with the following normalisation of the electron distribution function:

$$\int_0^\infty u^{1/2} f(u) du = 1 \quad (6)$$

so that the mean energy of the electrons is simply given by:

$$u_m = \frac{2}{3} \int_0^\infty u^{3/2} f(u) du. \quad (7)$$

Two often measured quantities in electron swarm experiments are the first Townsend coefficient, $\alpha_1/N = k_1/v_d$, and the characteristic energy $u_c = ED_\tau/v_d$.

3. Collision cross sections

In order to compute the electron distribution function, all the electron impact collision cross sections occurring in (1) must be known. For elastic processes, we have used the momentum transfer cross section given by Frost and Phelps (1964) except in the range 0.014 to 4 eV where we have adopted the data of Milloy *et al* (1977). This set of momentum transfer cross sections, derived from swarm experiments, is in good agreement with the recent measurements of the differential cross section for the elastic scattering (Srivastava *et al* 1981). For inelastic processes, we have used the set of collision cross sections previously established in paper I. This set of cross sections takes account of 40 individual excited levels and includes the three shells of ionisation. However, in the case of steady state discharges under study in the present work, the mean energy of the electrons is very small as compared to the ionisation potentials of the inner shells. As a result, the ionisation of the K and L shells may be neglected. Another difference between the REB-produced plasmas and the electric discharges lies in the energetically useful part of the inelastic collision cross sections. In REB-produced plasmas, the electron energies in a range extending from excitation thresholds up to about 20 eV are generally without importance, while in discharges this energetic domain plays the major role. Consequently, in the present work we have carefully analysed the shape of the inelastic cross sections at low impact energies.

The set of excitation cross sections developed in paper I is based, in part, upon the measurements of Chutjian and Cartwright (1981) which were performed at energies above 16 eV for the 4s levels and above 20 eV for the upper levels. In the near threshold range, the shape of the excitation function of the argon metastable states, 3P_0 and 3P_2 , was measured by Olmsted *et al* (1965) and by Pichanick and Simpson (1968). Time-of-flight experiments were also performed to give the excitation cross section of the metastable states in a relative scale (Lloyd *et al* 1972b) as well as in an absolute one (Borst 1974). All these determinations are in excellent agreement among themselves, so that the shape of the metastable excitation cross section at energies lower than 16 eV seems well established. However in all these experiments, the excitation of the 3P_2 and 3P_0 metastable states arises not only from direct processes but also via cascades from the upper levels. The mean branching ratio, $D(u)$, for the destruction of the 4p levels leading to the excitation of the metastable states 3P_0 and 3P_2 is given by:

$$D(u) = \sum_s \sum_p \left[T_{p,s} \left(Q_p(u) + \sum_d T_{d,p} Q_d(u) \right) \right] / \sum_p \left(Q_p(u) + \sum_d T_{d,p} Q_d(u) \right) \quad (8)$$

where the summation over s is over the 3P_0 and 3P_2 states, the summation over p is over the ten $4p$ and $4p'$ levels and the summation over d is over all the upper states. The $T_{j,i}$ are the individual branching ratios for processes corresponding to the destruction of an upper j -state leading to the formation of a lower i -state. The measurements of the excitation function as well as the time-of-flight experiments were performed at such low pressures that the collisional deactivation of the upper levels is negligible compared to their radiative de-excitation. Therefore the de-excitation branching ratios, $T_{j,i}$, are given by:

$$T_{j,i} = A_{j,i} / \sum_i A_{j,i} \quad (9)$$

where $A_{j,i}$ are the radiative transition probabilities. All the $A_{j,i}$ used in the present work are taken from Lilly (1976). In figure 1 we have shown the de-excitation branching ratios of the $4p$ levels. The full curve corresponds to the complete pattern in which the cascade from the upper levels onto the $4p$ levels is taken into account, whereas the broken curve results from a simplified scheme in which the $4p$ levels would be populated only via the direct excitation from the ground state (in this model $T_{d,p} = 0$). As we can see, the results are very sensitive to the inclusion of the upper cascade; when it is taken into account 60% of the $4p$ population is channelled onto the metastable states 3P_2 and 3P_0 and 40% decays towards the radiative levels 3P_1 and 1P_1 . These proportions are just opposite when the upper cascade is neglected. In a previous attempt to elaborate a set of collision cross sections, Eggarter (1975) obtained a D -value of 0.6 by accounting only for the $4p$ – $4s$ cascade and neglecting the upper one. His result was based on the measurement of the $4p$ excitation function by Zapesochnyi and Feltsan (1966). The agreement between the Eggarter D -value and our result shows that he implicitly accounted for the whole cascade. This result confirms the previous assumption (McConkey and Donaldson 1973) that the measurements of Zapesochnyi and Feltsan were seriously perturbed by cascade effects and do not reflect the excitation cross sections from the ground state. The importance of the cascades in the formation of the $4p$ levels appears also in figure 1 where the chain curve represents the branching ratio for the formation of these levels through cascades:

$$F(u) = \sum_p \sum_d T_{d,p} Q_d(u) / \sum_p \left(Q_p(u) + \sum_d T_{d,p} Q_d(u) \right). \quad (10)$$

For incident electron energy above 15 eV, the cascade from the upper levels is the main channel for the production of the $4p$ levels.

Once the cascade processes are properly handled, the apparent cross section, $Q_a(u)$, for the total production of the metastables 3P_2 and 3P_0 , can be calculated from the individual excitation cross sections:

$$Q_a(u) = \sum_s \left[Q_s(u) + \sum_p T_{p,s} \left(Q_p(u) + \sum_d T_{d,p} Q_d(u) \right) \right]. \quad (11)$$

The results are reported in figure 2 and compared to the experimental data of Borst. A very good agreement appears, except in a narrow range around 13 eV where our value is 25% lower than the experimental one. The peak in the measured cross section is due to the well known resonance effect correlated to the temporary formation of excited negative ions (Brunt *et al* 1976). This resonance feature is very sharp and may be neglected for our purpose. Apart from this effect, figure 2 shows that the individual cross sections derived in paper I for the excitation of the forbidden levels apply quite well at

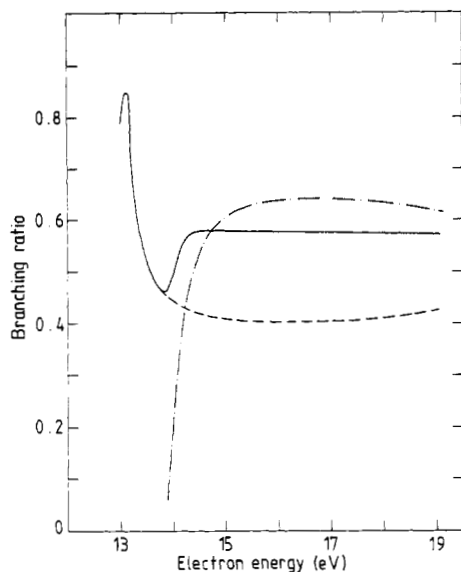


Figure 1. Branching ratios for the formation (chain curve) and the destruction (full and broken curves) of the 4p levels. The full curve corresponds to the inclusion of the upper cascades in the calculations while the broken curve results from a simplified model where they are neglected. The formation branching ratio accounts for cascades only.

low impacting energies, so that equations (6), (7) and table 2 of paper I can be used without modifications in the modelling of electrical discharges in argon.

Besides the work of Chutjian and Cartwright, few experimental determinations of the excitation cross sections of radiative states have appeared. McConkey and Donaldson (1973) reported an absolute determination of the excitation cross sections of the two first radiative levels. Their results, corrected from cascade effects, are in good agreement

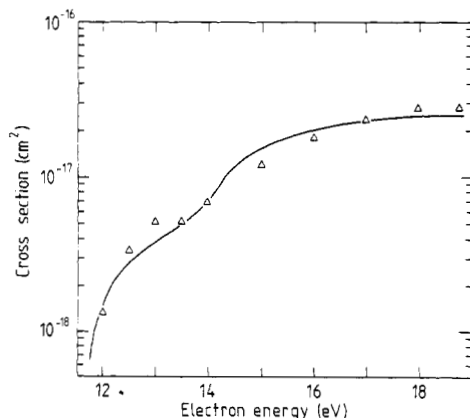


Figure 2. Apparent excitation cross section of the two metastable levels 3P_2 and 3P_0 as a function of the electron energy. —, present calculations; Δ , time-of-flight measurements of Borst.

with the value of Chutjian and Cartwright at 20 eV. They also agree, at energies above 50 eV, with the relative measurements of Lloyd *et al* (1972a) when these latter are normalised to the Born cross section at high incident energy. Compared to the McConkey and Donaldson results, the shape of the low-energy modifier, $B_j(E)$, proposed in paper I for the optically allowed transitions:

$$B_j(E) = [1 - U_j^{\alpha_j}]^{\beta_j} \quad (12)$$

where $U_j = W_j/E$, (W_j is the excitation threshold of level j), appears to underestimate the excitation cross section of the 3P_1 and 1P_1 levels at incident electron energies lower than 20 eV. As a result, we obtain good results with the following low-energy modifier:

$$B'_j(E) = 1 + A_j U_j + C_j U_j^2. \quad (13)$$

From a least-square analysis we determine $A_j = -1.275$, $C_j = 0.48$ for the 1P_1 state and $A_j = -1.035$, $C_j = 0.29$ for the 3P_1 state. At energies above 20 eV the two low-energy modifiers, $B_j(E)$ and $B'_j(E)$, give identical results, while at lower energies the differences in the resulting excitation cross sections are shown in figure 3. Indeed, we have verified that, as expected, the use of these improved low-energy modifiers for the two resonant levels has no noticeable consequences on the modelling of REB-produced plasmas (Bretagne *et al* 1986).

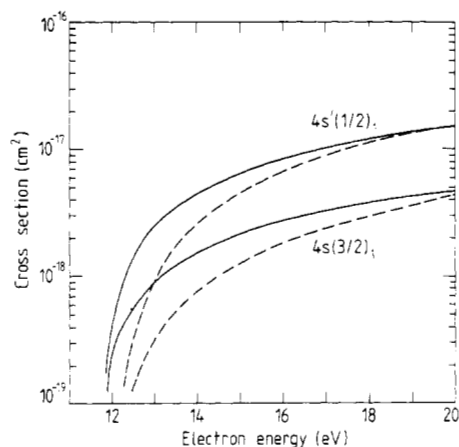


Figure 3. Excitation cross section of the radiative levels 3P_1 and 1P_1 at an incident electron energy lower than 20 eV. Full curves result from the improved low-energy modifier, equation (13) of the present work, while the broken curves result from equation (7) of paper I.

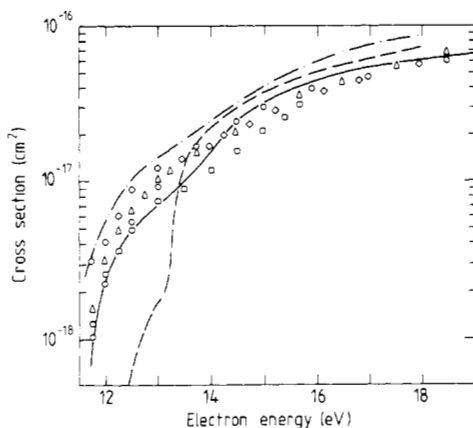


Figure 4. Total excitation cross section of argon versus the energy of incident electrons. Calculations: —, present work; ---, Peterson and Allen; — · —, Eggarter. Direct measurements: □, Schaper and Scheibner. Deductions from the first Townsend coefficient: ◇, Specht *et al* (1980); △, Sakai *et al* (1977); ○, Lagushenko and Maya.

We report, in figure 4, the total excitation cross section of argon which has been measured (Schaper and Scheibner 1969), calculated from a set of individual or composite cross sections (Peterson and Allen 1972, Eggarter 1975, present work) or deduced from the analysis of the first Townsend coefficient measured in swarm experiments (Jacob and Mangano 1976, Sakai *et al* 1977, Specht *et al* 1980, Lagushenko and Maya 1984).

For electron energies above 15 eV, the different total excitation cross sections are in fairly good agreement among themselves if we except the Eggarter values which are overestimated by about 50%. For incident energies just above the 4s excitation thresholds, some discrepancies appear. The results of Peterson and Allen largely underestimate the total cross section; this is expected because at energies lower than 13 eV, the metastable states 3P_0 and 3P_2 , not included in their set, play the dominant role in the total cross section. For energies lower than 13 eV, our results are in excellent agreement with the measurements of Schaper and Scheibner and with the determination of Lagushenko and Maya while the cross sections proposed by Specht and by Eggarter seem to be overestimated by a factor of two. The values of Sakai *et al* (1977), as well as those of Jacob and Mangano (not represented in figure 4 for clarity in the plotting) lies about 30% higher than ours.

4. Results and discussion

The electron distribution function has been calculated in the range $10^{-19} \leq E/N \leq 2 \times 10^{-15} \text{ V cm}^2$ and is reported in figure 5. For the reduced electric field, E/N , lower than $7 \times 10^{-17} \text{ V cm}^2$, the heating of the electrons by the electric field is so low that the shape of the distribution is determined only by elastic collisions and the distributions exhibit a dramatic decrease at energies much lower than the first excitation threshold. On the other hand, for E/N values greater than $7 \times 10^{-17} \text{ V cm}^2$, the electron distribution functions fall off rapidly only for electron energies above the excitation thresholds of the 3P_2 level as a result of inelastic collisions. No calculations were per-

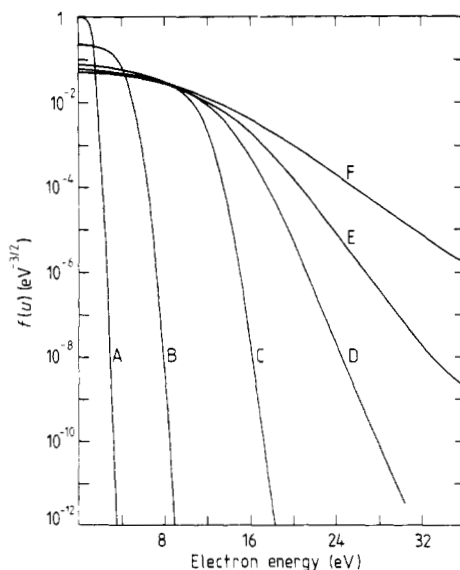


Figure 5. Electron energy distribution function in argon for different values of E/N : The curves are labelled as follows: A, $E/N = 10^{-18} \text{ V cm}^2$; B, $E/N = 10^{-17} \text{ V cm}^2$; C, $E/N = 10^{-16} \text{ V cm}^2$; D, $E/N = 5 \times 10^{-16} \text{ V cm}^2$; E, $E/N = 10^{-15} \text{ V cm}^2$; F, $E/N = 2 \times 10^{-15} \text{ V cm}^2$.

formed at E/N values higher than $2 \times 10^{-15} \text{ V cm}^2$, since at such high values of E/N the anisotropic part of the distribution becomes important and the two-term expansion of the Boltzmann equations seems no more valid as shown by Phelps and Pitchford (1985) in their study on N_2 . However, one can remark that in argon, the elastic cross section overcomes the total inelastic cross section by an order of magnitude up to an incident electron energy of about 20 eV. Therefore the two-term approximation should hold better than it does in molecular gases such as N_2 . For practical purposes, this limitation at $E/N = 2 \times 10^{-15} \text{ V cm}^2$ is not a drastic one because such a high value of the reduced field is not reached in excimer lasers.

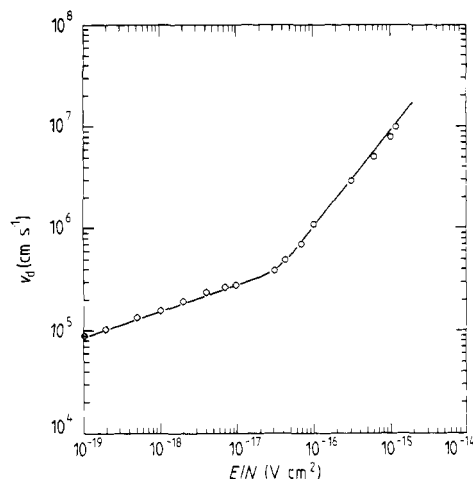


Figure 6. Electron drift velocity v_d as a function of E/N . —, present calculations based on experimental data of Robertson ($\diamond$) and of Kücükarpaci and Lucas ($\circ$).

The drift velocities are shown in figure 6, as a function of E/N . A very good agreement appears between our calculated values and the experimental data. In figure 6 we have only plotted the recent experimental data of Robertson (1977) and Kücükarpaci and Lucas (1981). For many years, much attention has been devoted to the determination of transport parameters and the results published prior to 1972, compiled by Dutton (1975), are not discussed here. Contrary to the drift velocity measurements, the experimental determinations of the characteristic energy, u_c , are very scarce. Apart from the measurements of Townsend (see Dutton 1975) the only recent experimental data are those of Milloy and Crompton (1977) for E/N lower than 10^{-18} V cm^2 and those of Lakshminarasimha and Lucas (1977) for $10^{-16} < E/N < 10^{-14} \text{ V cm}^2$. Our results, shown in figure 7, are in good agreement (within 10%) with the experimental data: the characteristic energy is roughly constant for E/N above $5 \times 10^{-17} \text{ V cm}^2$; however, the sharp discontinuity at $E/N = 5 \times 10^{-16} \text{ V cm}^2$ in the Lakshminarasimha and Lucas data is not reproduced in our calculations and requires further theoretical and experimental investigations. Also reported in figure 7 is the mean energy, u_m , of the electrons; for a Maxwellian electron distribution u_c and u_m are equal, so that their difference is indicative of the departure of the electron distribution from a Maxwellian one. As a result, it

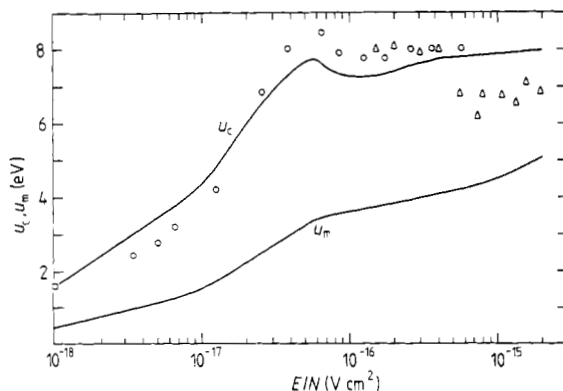


Figure 7. Characteristic energy u_c and mean energy u_m as a function of E/N . The full curves result from the present calculations while the symbols correspond to the following experimental values: $\circ$, Townsend (see Dutton 1975); $\square$, Milloy and Crompton; $\triangle$, Lakshminarasihma and Lucas.

appears in figure 7 that, in argon, the electron distribution function is highly non-Maxwellian for any E/N value, contrary to the electron distributions in molecular gases (Brunet and Vincent 1979) which become Maxwellian at high E/N values.

The first Townsend coefficient, α_1/N , is plotted in figure 8 versus the reduced electric field for E/N values greater than 10^{-16} V cm². α_1/N increases sharply with E/N and is a very sensitive function of the value of the total excitation cross section, especially for E/N values in the range $10^{-16} < E/N < 3 \times 10^{-16}$ V cm². At lower values, α_1/N vanishes and no determination is available. This property has been used by some authors (Jacob

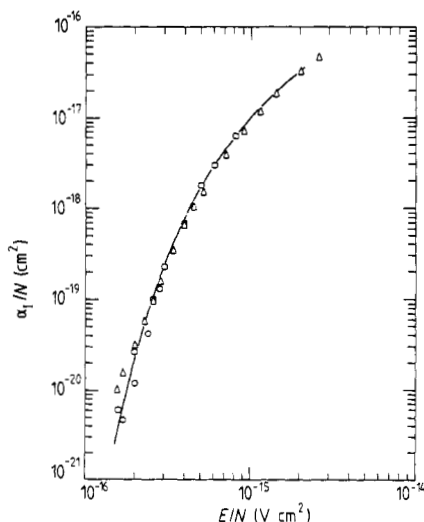


Figure 8. First Townsend coefficient α_1/N . The full curve is the present calculation. The symbols are the experimental data: $\triangle$, Kruithof and Penning (see Dutton 1975); $\circ$, Specht *et al* (1980); $\square$, Abdulla *et al* (1981).

and Mangano 1976, Sakai *et al* 1977, Specht *et al* 1980, Lagushenko and Maya 1984) to evaluate σ_{tot} from a fit between their predicted values of α_1/N and the experimental ones. Unfortunately, the experimental determinations of α_1/N for E/N lower than $3 \times 10^{-16} \text{ V cm}^2$ is very difficult because of spurious effects due to Penning reactions and surface photo-electron emission from the lowest excited states (Specht *et al* 1980). These difficulties give rise to a large scattering in the data and probably explain the spreads observed in figure 4 for the σ_{tot} values determined by this method. Our predicted values of α_1/N , based on a microscopic approach, are in excellent agreement with the most recent experimental data (Specht *et al* 1980, Abdulla *et al* 1981) in the whole investigated range of E/N .

Once the electron distribution function is known, the calculation of the excitation rate coefficients of the individual argon levels is performed through the use of (4), in

Table 1. Excitation rate coefficients ($\text{cm}^3 \text{s}^{-1}$) of argon levels for direct excitation from the ground state. Note $3.77(-13) = 3.77 \times 10^{-13}$.

State	$E/N (10^{-16} \text{ V cm}^2)$					
	0.5	1	3	7	10	20
4s(3/2) ₂	3.77(-13)	3.67(-12)	2.32(-11)	6.02(-11)	8.21(-11)	1.40(-10)
4s(3/2) ₁	7.03(-14)	8.40(-13)	7.25(-12)	2.48(-11)	3.94(-11)	9.06(-11)
4s'(1/2) ₀	4.37(-14)	5.26(-13)	3.90(-12)	1.07(-11)	1.49(-11)	2.61(-11)
4s'(1/2) ₁	1.18(-13)	1.84(-12)	1.91(-11)	7.08(-11)	1.16(-10)	2.80(-10)
4p(1/2) ₁	3.28(-16)	2.86(-14)	8.45(-13)	4.26(-12)	7.93(-12)	2.34(-11)
4p(5/2) ₃	2.30(-16)	2.89(-14)	1.09(-12)	6.09(-12)	1.22(-11)	4.05(-11)
4p(5/2) ₂	1.61(-16)	2.39(-14)	1.01(-12)	5.65(-12)	1.00(-11)	2.55(-11)
4p(3/2) ₁	1.69(-16)	2.79(-14)	1.23(-12)	6.64(-12)	1.14(-11)	2.72(-11)
4p(3/2) ₂	9.86(-17)	1.73(-14)	8.24(-13)	4.79(-12)	8.57(-12)	2.22(-11)
4p(1/2) ₀	1.84(-16)	3.99(-14)	2.12(-12)	1.23(-11)	2.14(-11)	5.24(-11)
4p'(3/2) ₁						
4p'(3/2) ₂	4.56(-17)	1.07(-14)	6.21(-13)	3.87(-12)	7.07(-12)	1.88(-11)
4p'(1/2) ₁	3.32(-17)	8.02(-15)	4.63(-13)	2.75(-12)	4.83(-12)	1.20(-11)
4p'(1/2) ₀	4.89(-17)	1.68(-14)	1.31(-12)	8.98(-12)	1.69(-11)	4.63(-11)
3d(1/2) ₀	2.55(-17)	2.00(-14)	2.73(-12)	2.06(-11)	3.80(-11)	9.77(-11)
3d(1/2) ₁	3.46(-21)	5.30(-18)	1.84(-15)	2.62(-14)	7.06(-14)	2.89(-13)
3d(3/2) ₂	1.68(-17)	1.55(-14)	2.37(-12)	1.86(-11)	3.46(-11)	8.99(-11)
3d(7/2) ₄	8.46(-18)	9.66(-15)	1.72(-12)	1.41(-11)	2.66(-11)	7.02(-11)
3d(7/2) ₃	1.97(-18)	2.52(-15)	5.00(-13)	4.40(-12)	8.73(-12)	2.47(-11)
3d(5/2) ₂	1.81(-18)	2.70(-15)	5.96(-13)	5.42(-12)	1.09(-11)	3.11(-11)
5s(3/2) ₂						
5s(3/2) ₁	2.06(-20)	6.20(-17)	3.40(-14)	5.64(-13)	1.61(-12)	6.87(-12)
3d(3/2) ₁	2.63(-20)	9.63(-17)	6.00(-14)	1.04(-12)	3.00(-12)	1.30(-11)
3d'(5/2) ₂	1.09(-18)	2.50(-15)	7.12(-13)	6.73(-12)	1.34(-11)	3.70(-11)
3d'(3/2) ₂	5.30(-19)	1.32(-15)	4.14(-13)	4.18(-12)	8.69(-12)	2.58(-11)
3d'(5/2) ₃						
5s(1/2) ₀	2.85(-21)	1.48(-17)	1.14(-14)	2.12(-13)	6.29(-13)	2.79(-12)
5s'(1/2) ₁						
3d'(3/2) ₁	1.75(-20)	1.08(-16)	9.27(-14)	1.78(-12)	5.36(-12)	2.40(-11)
a	5.19(-23)	7.82(-18)	2.09(-14)	5.82(-13)	2.01(-12)	9.89(-12)
b	0	3.17(-18)	1.75(-14)	6.18(-13)	2.32(-12)	1.21(-11)
c	0	5.62(-19)	5.81(-15)	2.51(-13)	1.02(-12)	5.55(-12)
d	0	3.54(-19)	8.73(-15)	4.93(-13)	2.21(-12)	1.28(-11)

Table 2. Excitation rate coefficients ($\text{cm}^3 \text{s}^{-1}$) of argon levels for cascades from the upper levels.

State	$E/N (10^{-16} \text{ V cm}^2)$					
	0.5	1	3	7	10	20
$4s(3/2)_2$	6.81(-16)	1.15(-13)	8.84(-12)	6.31(-11)	1.19(-10)	3.21(-10)
$4s(3/2)_1$	3.90(-16)	7.15(-14)	4.77(-12)	3.26(-11)	6.03(-11)	1.59(-10)
$4s'(1/2)_0$	1.14(-16)	2.48(-14)	1.66(-12)	1.11(-11)	2.03(-11)	5.22(-11)
$4s'(1/2)_1$	1.69(-16)	4.46(-14)	3.30(-12)	2.29(-11)	4.27(-11)	1.15(-10)
$4p(1/2)_1$	3.50(-17)	2.91(-14)	4.17(-12)	3.20(-11)	5.92(-11)	1.53(-10)
$4p(5/2)_3$	9.36(-18)	1.07(-14)	1.91(-12)	1.57(-11)	2.98(-11)	7.92(-11)
$4p(5/2)_2$	2.38(-18)	3.25(-15)	6.84(-13)	6.13(-12)	1.22(-11)	3.47(-11)
$4p(3/2)_1$	3.30(-18)	3.46(-15)	6.27(-13)	5.38(-12)	1.06(-11)	2.95(-11)
$4p(3/2)_2$	2.97(-18)	2.84(-15)	4.62(-13)	3.75(-12)	7.15(-12)	1.92(-11)
$4p(1/2)_0$	1.05(-18)	2.12(-15)	5.82(-13)	5.46(-12)	1.08(-11)	3.00(-11)
$4p'(3/2)_1$						
$4p'(3/2)_2$	1.10(-18)	1.67(-15)	4.16(-13)	3.94(-12)	7.96(-12)	2.29(-11)
$4p'(1/2)_1$	1.03(-18)	1.08(-15)	2.13(-13)	1.91(-12)	3.79(-12)	1.07(-11)
$4p'(1/2)_0$	3.88(-22)	2.25(-18)	1.88(-15)	3.60(-14)	1.08(-13)	4.81(-13)

which the $Q_j(u)$ are given by (5)–(7) and tables 1 and 2 of paper I (indeed, for the $4s(3/2)_1$ and $4s'(1/2)_1$ levels we use (13) of the present paper rather than (7) of paper I). In order to save some computational time, we have grouped together some levels lying above 14.5 eV. In practice, these levels suffer a very rapid associative ionisation and play only a minor role in the kinetic scheme of excimer lasers. Namely, we have gathered the levels $4d(1/2)_1$, $6s(3/2)_1$ and $4d(3/2)_1$ in an effective level labelled 'a', the levels $4d'(3/2)_1$, $6s'(1/2)_1$, $5d(1/2)_1$, $7s(3/2)_1$ and $5d(3/2)_1$ in a level 'b', the levels $6d(1/2)_1$, $5d'(3/2)_1$, $7s'(1/2)_1$, $8s(3/2)_1$ and $6d(3/2)_1$ in a level 'c' and all the allowed levels above the $7d(1/2)_1$ in a level 'd'. All the atomic parameters describing these levels are constructed from the individual atomic parameters through statistical rules. The excitation rate coefficients of all the levels studied in the present work are given in table 1 for some values of the reduced field. The excitation rate coefficients of some of these levels have been measured very recently with a drift tube technique (Tachibana 1985). To our knowledge, apart from these experiments, only the excitation rate coefficient of the $4s(3/2)_2$ level has been previously measured in a low-pressure argon positive column (Ferreira and Ricard 1983). But in these two kinds of experiments, besides their direct excitation from the ground state, the argon levels are also produced by cascades from the upper levels, thus we must calculate the cascade excitation rate coefficients before making a comparison. From (4) and (11) in which the first summation over s is omitted, one easily obtains the cascade excitation rate coefficient of a j -level:

$$k_j^c(E/N) = \sum_p T_{p,s} \left(k_p(E/N) + \sum_d T_{d,p} k_d(E/N) \right). \quad (14)$$

The results of (14) are given in table 2 for some E/N values. Indeed, the cascade excitation rate coefficients are calculated only for the $4s$ and $4p$ levels, they are zero for the upper states. The comparison of the direct excitation processes and the cascade ones reveals some interesting features. For reduced fields lower than 10^{-16} V cm^2 , the excitation of the $4s$ and $4p$ levels by the cascade is low, less than 10%, except for the $4p(5/2)_3$ and $4p(1/2)_1$ levels for which the two channels have the same magnitude. The

enhancement of the reduced electric field produces an increase of the cascade excitation faster than that of the direct excitation. As a result, for most of the levels, the two excitation channels compare for an E/N value of about $7 \times 10^{-16} \text{ V cm}^2$, while at higher values the production due to the cascade overcomes the direct excitation from the ground state. The only two exceptions to this general behaviour are the $4s'(1/2)_1$ and $4p'(1/2)_0$ levels; for these levels the direct excitation remains the main production process at any E/N value. This effect is particularly sensitive for the $4p'(1/2)_0$ level for which the cascade contributes for less than 1% of its total excitation.

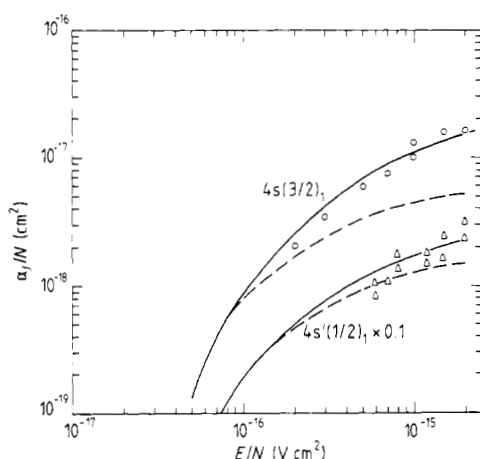


Figure 9. Excitation rate coefficient α_i/N versus E/N for the two first resonant levels of argon. The values of the $4s'(1/2)_1$ level have been reduced by a factor of 10. The full curve is the present calculation including both the direct excitation and the cascade from upper levels; the broken curve is the present calculation for only the direct excitation. The symbols are the measurements of Tachibana.

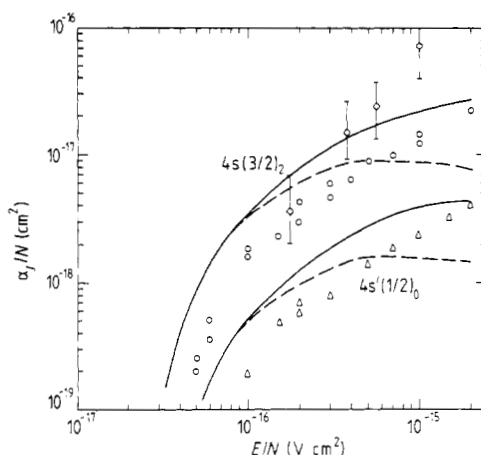


Figure 10. Excitation rate coefficient α_i/N versus E/N for the two first metastable levels of argon. The symbols labels are the same as in figure 9; in addition the symbols with error bars are the measurements of Ferreira and Ricard for the $4s(3/2)_2$ level.

The comparison with the experimental results is shown in figures 9–11, where we have plotted, versus E/N , the excitation rate coefficients in units of cm^2 : $\alpha_i/N = k_i/v_d$. The full curve represents our calculated total excitation rate coefficients while the broken curves correspond to the direct processes alone. The agreement between our predicted values and the measurements of Tachibana (1985) is very satisfactory except for the two metastable levels, $4s(3/2)_2$ and $4s'(1/2)_0$, for which our values lie above those of Tachibana by a factor of about two. However, our calculated excitation rate coefficients for the $4s(3/2)_2$ level are in better agreement with the measurements of Ferreira and Ricard (1983). Figures 9–11 emphasise that for the most of the levels, the cascades are the main channel of formation at high E/N values. The noticeable exceptions are the $4s'(1/2)_1$ and $4p'(1/2)_0$ levels. For this latter level the broken and the full curves are superposed on the whole E/N range.

Figure 12 gives the percentage of power transferred into the various elastic or inelastic processes as a function of E/N . In order to simplify this figure we have only shown a few

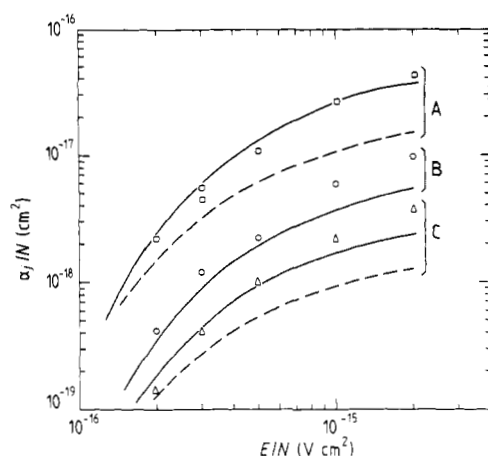


Figure 11. Excitation rate coefficient α_i/N versus E/N for two individual levels of the 4p configuration and for all the ten 4p levels (A). The symbols are the same as in figure 9. For the $4p'(1/2)_0$ level (B) the cascade excitation is negligible and the full and broken curves are superposed; this level has been multiplied by a factor of two. (C) represents the level $4p(3/2)_2$.

groups of discrete levels according to their configuration 4s, 4p and upper states. For practical purposes these fractional energy losses are of interest because they are directly related to the laser efficiency. Figure 12 clearly shows that in discharge-pumped lasers, the production efficiencies of the various levels are very sensitive functions of the applied electric field. For example, the best efficiency for the pumping of the 4s levels is obtained with an applied field of about 10^{-16} V cm². Indeed, such a value corresponds to that used to energise the excimer lasers. At this E/N value the fractional power transferred into each of the 4s levels is 40% for the level $4s(3/2)_2$, 9% for the level $4s(3/2)_1$, 6% for the level $4s'(1/2)_0$ and 20% for the level $4s'(1/2)_1$.

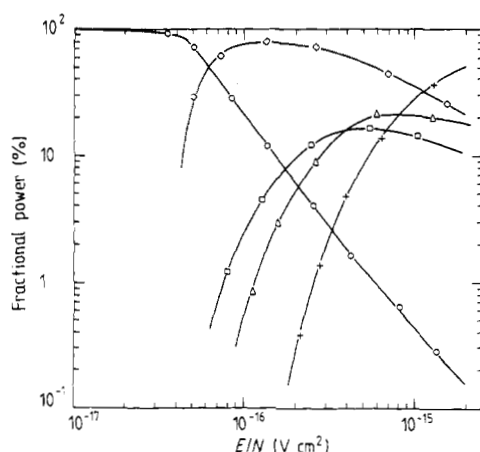


Figure 12. Fractional power transfer into different processes as a function of E/N . $\circ$, elastic collision; +, ionisation of argon; $\diamond$, excitation of the 4s levels; $\square$, excitation of the 4p levels; $\triangle$, excitation of the upper levels.

5. Conclusion

The main purpose of the present paper was to investigate whether the detailed set of electron–argon collision cross sections previously proposed (Bretagne *et al* 1986) could be effectively used to calculate the excitation rate coefficients which are the necessary inputs of any realistic modelling of discharge-pumped excimer lasers. The predicted swarm parameters are in excellent agreement with the best available experimental data. In the present work based on a microscopic approach, these swarm parameters are only used as a check of the calculations and never as a guideline for adjustments in the collision cross sections through fitting procedures. This latter macroscopic approach, frequently used, can only give an estimate of the total excitation cross section but is unable to give knowledge of the individual processes occurring in the discharge.

For about 30 excited levels we have calculated the excitation rate coefficients for the direct excitation from the ground state and for cascades from upper levels. For the few levels for which measurements are available, a good agreement is observed between the experimental data and our predicted values. At an E/N value of 10^{-16} V cm², commonly used to energise the excimer lasers, our calculations show that 75% of the input power is transferred into the 4s levels and only 3% into the upper levels; the remaining energy is consumed in elastic collisions. As a result, the cascades from upper levels play only a minor role in the laser kinetics. Moreover, the present analysis indicates that, in the modelling of high current discharge-pumped lasers, the main process which should be carefully studied in the near future is the excitation and the ionisation of each of the 4s levels.

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